Methodology

The manufacturing process involves electronics assembly and electromechanical hardware assembly. In an electronic assembly, all components are soldered on pre-designed PCBs. The assembled PCBs are tested for desired performance.

Electromechanical components such as connectors, sockets, switches and LED displays are assembled as per the design of the panel of the equipment. Subsequently, electronic and electromechanical assemblies are assembled to form a compact unit and encased in a metallic or plastic cabin. The final product is then calibrated and tested as per relevant design specifications and customer requirements.

TABLE I
FIXED CAPITAL

LAND AND BUILDING

Description	Details
Built-up area	150sqm
Office, stores	50sqm
Assembly and testing	100sqm
Rent (₹ per month)	10,000

MACHINERY AND EQUIPMENT

Description	Indian/Imported	Quantity	Value (₹)
Drilling machine	Indian	1	6,000
Grinder (portable)	Indian	1	5,000
Oscilloscope (50MHz)	Indian	1	50,000
LCR-Q meter	Indian	1	15,000
Power supply (0-30V, 3A)	Indian	1	20,000
Insulation tester	Indian	1	5,000
Testing setup (voltmeter, ammeter, wattmeter, etc)	Indian	1	40,000
Digital multimeter	Indian	2	20,000
Analogue multimeter	Indian	2	3,000
Personal computer with UPS and printer	Indian	1	60,000
Electrification and installation charges @ 10% of the total above			22,400
Temperature-controlled soldering stations, tools, jigs, fixtures, electronic screw drivers, etc			50,000
Office equipment and furniture			30,000
Pre-operative expenses			20,000
Total Fixed Capital			**346,400**

as insulators in a variety of electronic components.

Major equipment and machinery needed for this business are as follows:

1. Drilling machine
2. Grinder (portable)
3. Oscilloscope (50MHz)
4. LCR-Q meter
5. Power supply (0-30V, 3A)
6. Insulation tester
7. Testing setup (consisting of voltmeter, ammeter, wattmeter, etc)
8. Digital multimeter
9. Analogue multimeter
10. Temperature-controlled soldering stations, tools, etc
11. Jigs, fixtures, electronic screwdrivers, etc
12. Office equipment and furniture

Pollution control practices

The government of India has pledged to give utmost importance to control environmental pollution. To support this drive, small-scale entrepreneurs are required to have an environmental-friendly attitude and adopt pollution-control measures by process modification and technology substitution.

In the business of electronic counter manufacturing, a great amount of plastic is used, most of which is not recyclable. The electronics industry uses chlorofluorocarbons (CFC), carbon tetrachloride and methyl chloroform for cleaning PCBs after assembly, to remove flux residues left after soldering, and various kinds of foams for packaging. To reduce harmful chemicals emission in the atmosphere, the steps given below can be followed.

Fumes and gases are released during soldering, which are harmful to people as well as the environment. Alternative technologies may be used to phase out existing polluting technologies.

Numerous new fluxes have been developed, which contain two to ten per cent solids as opposed to the traditional 15 to 33 per cent. To remove flux residues from PCBs, alternative solvents could replace CFC-113 and methyl chloroform.

Chlorinated solvents such as trichloroethylene, perchloroethylene and methylene chloride have been used as effective cleaners for many years. Organic solvents like ketones and alcohols are effective in removing both solder fluxes and polar contaminants.

Business economics

An indicative analysis is given on the following basis and presumptions:

1. Basis for production capacity calculation has been taken on a single shift basis, on 75 per cent efficiency. Therefore the total number of counters produced per month is 300 only, and per year 3600.

2. Maximum capacity utilisation is done on single shift basis, for 300 days in a year. During the first and second years of operation, capacity utilisation is assumed to be 60 per cent and 80 per cent, respectively. The unit is expected to achieve full capacity utilisation third year onwards.

3. Salaries and wages, cost of raw materials, utilities, rent and so on are based on the prevailing rates in and around Thrissur. These cost factors may vary with time and location.

4. Interest on term loan and working capital has been taken @ 16% on an average. This rate may vary depending on the policy of different financial institutions/agencies from time to time.

5. Cost of machinery and equipment refer to a particular make/model. Prices are approximate.

6. Break-even point percentage indicated is of full-capacity utilisation.

7. Project preparation costs and the like, whenever required, can be con-